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Task Description:

NASA GRC supports Integrated Medical Model (IMM) by developing simulation based models that forecast medical event likelihood from first principle and root cause information. The specific models to be developed to inform IMM are typically assigned and prioritized by the Exploration Medical Capabilities Element and the science and clinical team supporting the IMM. In general this effort utilizes a modified simulation probabilistic risk analysis (Simulation PRA) approach where deterministic models are used to integrate lower level observed phenomena to assess the likelihood of higher level clinical events. Since this approach is based on PRA, it is scenario dependant and must be carefully constructed to include all critical information. There are 4 Primary Projects in FY2010.
Sleep Disruption Medical Intervention Forecasting (SDMIF) Tool

During International Space Station (ISS) and shuttle missions, difficulties with sleep affect more than half of all US crews. Mitigation strategies to help astronauts cope with the challenges of disrupted sleep patterns can negatively impact both mission planning and vehicle design. The methods for addressing known detrimental impacts for some mission scenarios may have a substantial impact on vehicle specific consumable mass or volume or on the mission timeline. As part of the Integrated Medical Model (IMM) task, NASA Glenn Research Center (GRC) is leading the development of a Monte Carlo based forecasting tool designed to determine the consumables required to address risks related to sleep disruption. The model currently focuses on the International Space Station and uses an algorithm that assembles representative mission schedules and feeds this into a well validated model that predicts relative levels of performance, and need for sleep (SAFTE Model, IBR Inc). Correlation of the resulting output to self-diagnosed needs for hypnotics, stimulants, and other pharmaceutical countermeasures, allows prediction of pharmaceutical use and the uncertainty of the specified prediction.

Head (and Neck) Injury Risk Module (HIRM)

The possibility of head and neck injury in space is a concern due to the negative impact it could have on a mission. The Head and neck injury risk module (HIRM) is under development at the NASA Glenn Research Center. The HIRM is a statistical simulation tool that attempts to quantify the probability of Head and neck injuries for particular activities or events during space exploration missions. The tool also seeks to distinguish between soft and hard tissue modalities.

Abdominal Injury Module (AIM)

The possibility of abdominal injury in space is a concern due to the negative impact it could have on a mission. The Abdominal injury risk module (AIM) is under development at the NASA Glenn Research Center. The AIM is a statistical simulation tool that attempts to quantify the probability of abdominal injuries for particular activities or events during space exploration missions. The tool distinguishes injuries on clinical severity levels and between solid and hollow organs. The basis for the tool is the same conceptual framework as utilized for the GRC Bone fracture risk module.

Bayesian Medical Event Predictions (BMEP)

Several medical conditions have clear earth analogs, but the rate of the occurrence in the astronaut population during space travel is not well understood due limited occurrences. Using numerical Bayesian techniques, the NASA Glenn Research Center is developing a series of analyses to provide the IMM with the mean likelihood and estimated uncertainty of the occurrences specific medical conditions that are prominent in the general population and may appear in a specific individual during the time frame of a space mission (weeks to years). These can include, glaucoma (closed angle), stroke, seizure, and other similar conditions identified by the IMM team.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

As this research is targeted to assessing risk of medical incidences to astronauts while in-flight, there are potential applications in the general clinical setting. In concept, because these models depend on well accepted biomechanical, physiological and other performance models, the approaches we have taken can be modified to address medical likelihoods for specific individuals or groups provided that sufficient scenarios and descriptive data can be provided. As an example, the model for predicting the likelihood of bone fracture can be modified to assess one's yearly likelihood of fracture given details regarding BMD (bone mineral density) and planned "potentially hazardous" activities like skiing or mountain biking. Similarly, the renal stone model can be used to assess how treatment of a different malady may affect the likelihood of renal colic. These and other applications could arise from our efforts provided the models are advanced to include these applications.

Task Progress for 2007 - 2010
Bone Fracture Risk Model (BFxRM)

The BFxRM mathematically models the probability of bone fracture during specific astronaut activities during space exploration missions. We focused on events that represent severe consequences for an exploration mission, specifically that of spinal fracture resulting from a routine task (lifting a heavy object up to 60 kg), or a spinal, femoral or wrist fracture due to an accidental fall or an intentional jump from 1-2 m. We validated the biomechanical and bone fracture models against 18 terrestrial studies of ground reaction forces, skeletal loading, fracture risk, and fracture incidence. After validation in a terrestrial setting, we predicted fracture risk associated with reference missions to the moon and Mars that represented crew activities on the surface. The model was extended to address fracture likelihood for 6-month ISS missions.

Generally, femoral neck fracture probability due to a fall to the side during an EVA (extravehicular activity) is very low even during a long-duration Mars mission, possibly due to the effective use of spacesuit padding. The static lift model represents an astronaut slowly picking up or holding a heavy object (up to 60 kg, mean 20 kg) on Mars or the moon. The estimated probability of spinal fracture during an Intra-Vehicular Activity (IVA) in this scenario is similarly low, even at the most punishing flexion angle. However, the added weight and compromised bone integrity during an EVA on a Mars long-duration mission leads to a mean fracture probability of slightly less than 1%. Considering the disastrous consequences to the crew and the mission, this scenario warrants further consideration. The natural response to a fall or other perceived impact is to extend a hand to absorb some of the impact. This kind of active response by the astronaut could protect the hip, but lead to a wrist fracture, which we found to be the most likely type of fracture. While this type of injury is unlikely to be fatal, it can cause great discomfort and jeopardize mission success. The 95th percentile for Mars missions ranges from 5-8%, which is not insignificant. Gender-specific differences in mean wrist fracture

probability are greater than that for fracture of the hip or lumbar spine.

After modification of the BFxRM to utilize loading and statistical event data related to ISS injury modalities, a series of simulations were performed to assess likelihood of wrist fracture during a 6-month increment were performed. The result show that the mean likelihood is 3 orders of magnitude below that estimated for planetary missions and the 95th-percentile is not significantly different than the mean.

Of the model parameters related to EVA scenarios the force attenuation characteristic of the astronaut EVA suit represented a large epistemic uncertainty. This finding led to the follow on Suit Impact load Attenuation Study (SILAS), an investigation to quantify the base level of fracture protection that the EVA suit provides the astronaut. The data collected in this analog was used to inform the BFxRM, reducing the uncertainty when predicting the probability of fracture for a fall to the side impact event in lunar and Martian gravity.

For all suit pressures, the attenuation increased as the offset increased and varied over a range of 51 – 94%, 29 – 96% and 17 – 89% for the 5, 10 and 15 cm drop heights, respectively. The largest variation in attenuation occurred at offset values between 0.1 – 1.5 cm, where the mean attenuation values for these offsets were 69 ± 15%, 49 ± 22% and 35 ± 18%, for mean impact reference forces of 4827, 6400 and 8467 N, respectively. The mean attenuation values for offset values of 2.8 – 5.3 cm were 93 ± 2%, 94 ± 1% and 84 ± 5% for the same mean impact reference forces.

Attenuation values varied less due to differences in operating pressure and loading than with offset. When the attenuation values were averaged over the operating pressures for each fall height and offset combination, the standard deviations of these mean attenuation values ranged from 3 – 11%. When the attenuation values were averaged over the fall height for each operating pressure and offset combination, the standard deviations of these mean attenuation values ranged from 4 – 22%. When the attenuation values were averaged over the offsets for each operating pressure and fall height combination, the standard deviations of these mean attenuation values ranged from 14 – 32%.

The reduction in uncertainty in the BFxRM comes about because a higher fidelity model was created using the SILAS data. When only considering a suit operating pressure of 33 kPa, since it is the current operating pressure of the ISS EMU (extravehicular mobility unity) suit and the estimated operating pressure of the new lunar and Martian EVA suits, the mean attenuation value found in the SILAS study was 72 ± 27%, across all offset and loading values. This is comparable to the hip protector attenuation value of 73 ± 17.5% used in the first version of the BFxRM. However, instead of using a single, broad attenuation distribution, the SILAS data allowed the attenuation to be modeled as a function of impact loading during the fall and the size of the gap between the suit and the thigh. The confidence interval of the distribution used to describe the attenuation value was much smaller once it was focused using offset and loading values that were representative of the EVA conditions.

The timeline for the development of the BFxRM to its current state is as follows:

• 11/2005 - GRC asked to quantify bone fracture risk in astronauts

o Multidisciplinary Team from NASA-GRC and the Cleveland Clinic formed

• 4/2006 – Initial model concept presented to HRP-PRA working group for go ahead

• 2/2007 – Presented preliminary model results on lunar and Martian mission scenarios to Human Research Program (HRP)

• 3/2007 – 1/2008

o Added vertebral fracture predictions

o Delivered data to IMM

o Presented at multiple risk venues for PRA and risk expert feedback

• 1/2008 – Analyzed long duration space flight contribution to “near term” post flight fracture risk

• 5/2008 – Model used to enhance FRAX predictions of risk from off nominal Orion landings

• 7/2008 – Extended to ISS risks and to address wrist fracture

o validation completed 1/2009

• 11/2009 - Published results in Ann Biomed Eng. and NASA HRP evidence book

• Currently – Updating the post flight study in preparation for publication

Task Progress:

Foundational Assessment of Renal Stones in Space (FARiS)

Given that astronauts could experience stone promoting urine chemistry, potentially exacerbated by hypogravity exposure, there is the possibility that astronauts will develop clinically significant renal stones in flight. However, with the exception of terrestrial experience and anecdotal flight evidence, there is little capability to estimate the true likelihood of stone formation in space. This investigation seeks to develop a model of the urinary system which allows the incorporation of a variety of physiological and possibly hereditary factors into the calculation of risk of stone formation and to use this model to estimate the effect of water intake on the probability of renal stones in the astronaut population. Currently a proof of concept model has been completed using a number of simplifying assumptions. The predictions of the current model as-implemented imply a relatively large increase in the number of stone nidus can form in the tubules due heterogeneous growth alone. Further development of this module will seek to include agglomeration, wall effects and more complex urine chemistry to determine the probability change in clinical stone occurrence.

The predictions of the current model imply a relatively large number of relatively large sized nidus could form due heterogeneous growth alone with a complex dependence on water intake levels. By assuming a mean rate of uric acid crystal nucleation of 0.004 per nephron per minute as the seed surface for heterogeneous growth, a 50% correlation between uric acid nucleation and active heterogeneous stone growth in a nephron and assuming that only about 15% of the total number (~2x10(6)) of nephrons participate in stone-formation, estimates of the probability of large radii (order of the nephron radii) can be determined. For the 2.5 L/day case, probability of a nephron producing a nidus with radii of the nephron is 3.9E-14 per nephron per minute, corresponding to a probability of 2.4E-11 probability of a maximum

radii nidus per minute for the entire nephron bed. The 2.0 L/day case produces a 6.7E-06 probability of forming per minute per nephron – a substantially higher formation rate- corresponding to a net probability of about 4.0E-03 probability of a maximum radii nidus per minute for the entire nephron bed. The total stone formation rate, including sub- critical and blocked stones, is comparable in the two cases, although the differences in probability shows that water intake has a profound effect on nidus growth and implies a substantial influence on the subsequent growth and agglomeration.

A merit review of this model outlined several positive aspects of the approach, but also detailed a number of limitations. These limitations included the need for more precise information on the local physiology and the how certain data was interpreted. This was especially true of the limitation of using a normal distribution for all data sets in the model. Issues with utilizing concentrations of stone promoters and stone inhibitors were also identified. These will be addressed over the next three years in a follow on study beginning in Sept of 2010. The timeline for the development of the preliminary renal stone model was as follows:

• 6/2007 – Asked to look into quantifying the likelihood of renal stone formation by the ExMC (Exploration Medical Capability) element

• 8/2007 – Provided an initial Bayesian analysis defining likelihood based off US population and astronaut post-flight incidence data.

• 6/2008 – Asked to address impact change in water intake has on renal stone formation by the ExMC element

• 8/2008 – Developed conceptual model that combined deterministic predictions of renal crystal growth with probabilistic models of renal function

• 9/2008 – Contracted N&R engineering and National Center for Space Exploration and Research (NCSER) to implement conceptual model

• 12/2008 – Model constructed to analyze nidus formation rates at the end of the renal tubal completed

• 1/2009 – Interim report completed – project put on hold by ExMC element until merit review

• 11/2009 – Merit review with SMES to assess project feasibility

• 2/2010 – GRC team response to reviewer comments completed and submitted to SMO (Science Management Office)

• 5/2010 – 2nd phase model development started.

Sleep Disruption Medical Intervention Forecasting Tool (SDMIF)

Developed in partnership with collaborators at Air Force Research Laboratory (AFRL) and IBR, Inc., the SDMIF tool is designed to estimate the occurrence and impact of sleep disturbance with regard to mission planning and the need for medical interventions. This modeling tool integrates potential mission schedule changes with current knowledge of how such changes could impact astronaut sleep quality and quantity. By considering the current clinical practices in prescribing medical interventions to address disrupted sleep conditions, the tool forecasts the amount of sleep aids and stimulants required per astronaut, per mission to help mitigate reductions in alertness and performance. This module has the potential to assist International Space Station (ISS) planners in identifying the optimal quantity of sleep-promoting and alertness-enhancing pharmaceuticals needed to ensure unimpaired astronaut performance during future space flight operations.

At present the model implementation has been completed and function verified using subject matter expert data to inform the model. We are confident that the model is performing as designed and detailed in the conceptual model. At present, the lack of access to specific data sampling of astronaut sleep med use during an ISS mission prevents finalizing and validating the SDMIF. We are currently collaborating with Dr. Laura Barger who is currently completing an inflight study addressing both sleep quality and the use of sleep meds. We are presently completing obtaining permission from the study participants to use their data in the SDMIF. The current time line for the development of the tool has been as follows:

• 2/2007 – Asked to look into how PRA could assist BHP analysis

o Generated preliminary ideas for review

• 10/2007 - Given task to estimate the occurrence of insomnia/circadian disturbance in spaceflight

• 4/2008 - Converted task metric to the rate of sleep aid use as surrogate metric for significant insomnia

• 9/2008 Presented preliminary results at the 2008 International Astronautical Congress (IAC)

• 10/2008 SME (subject matter experts) Panel merit review of conceptual model and preliminary results

o Identified beneficial in-progress studies that can provide data

• 10/2009 Completed all development and panel recommendations

• Awaiting completion of data sharing agreement with identified studies

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